CrunchRadar

APU Cloud Innovation Hackathon 2026

INTELLIGENT AUTOMATION · COHORT APD2F2606SE

Spot the crunch before students hit it.

An AI agent that reads every deadline and assignment brief, works out the real hours, and moves deadlines out of overloaded weeks. A human approves every change.

Scan to try the live demo
joseph-mayala.github.io/crunchradar

THE PROBLEM

Deadlines pile up in the same weeks.

- **Every lecturer sets deadlines alone.** Nobody sees the whole cohort's calendar, so 4–5 assessments land in one week.
- **Hidden tasks.** Presentations and vivas sit inside the briefs but never show on the Moodle calendar.
- **Festive weeks.** Deadlines fall during Chinese New Year or Hari Raya.
- **No time for feedback.** The next assignment is due before marks on the last one come back.

THE RULE WE MEASURE AGAINST

How much is too much?

21 credits × 40 h (Student Learning Time)	840 h
Spread over a 14-week semester	≈ 60 h / week
Minus classes	- 20 h
Minus revision and reading	- 12 h
Assignment work limit	**28 h / week**

Malaysia's MQA rule: 1 credit = 40 hours of student effort. CrunchRadar counts hours of work, not just the number of deadlines.

HOW IT WORKS

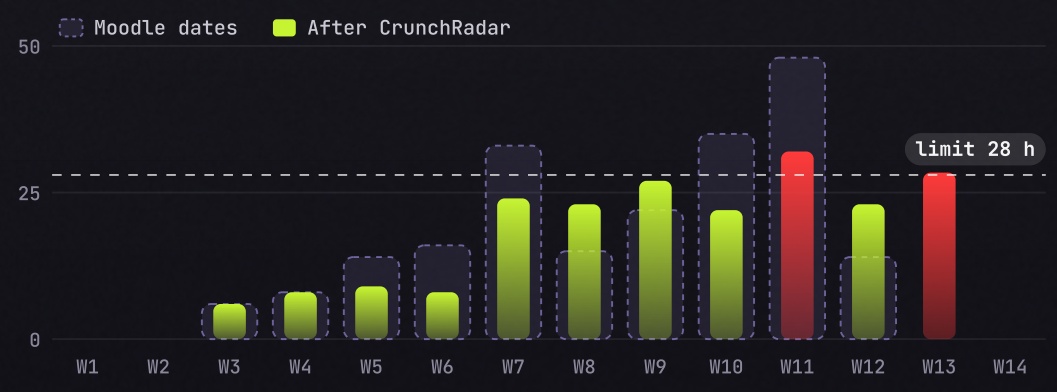
The Monday crunch check runs by itself, every week.

1. **Wake up** — Starts on its own every Monday at 8:00 AM. `Schedule trigger`
2. **Read** — Pulls every deadline and every assignment brief. `Calendar · Drive`
3. **Think** — Estimates hours per task and finds hidden tasks in briefs. `Gemini`
4. **Decide** — Any week over 28 h? Proposes moving a deadline 1–2 weeks later. `Condition`
5. **Ask a human** — Programme leader approves or rejects with one click. `Approval`
6. **Act** — Emails the lecturer, updates students, checks Moodle was changed. `Outlook`

↻ Repeats every week, all semester, with nobody logged in

RESULT IN THE DEMO COHORT

4 approved moves flatten the peak.



48 h → 32 h peak week

3 → 2 crunch weeks

2 hidden tasks found

WHO IT HELPS

Three people, one calendar.

- **Students** — A Monday email with this week's hours, what to start now, a "marks per hour" priority list, and an anonymous "this week is too much" button.
- **Lecturers** — Check a new assignment against the cohort's load before publishing it. Extension requests arrive with a drafted decision.
- **Programme leader** — Sees every crunch week on one heatmap and approves fixes in one click. Unfixable weeks are escalated with options.

BUILT FOR GEMINI ENTERPRISE WITH

5 workflow agents · Google Calendar · Google Drive · Google Forms · Microsoft Outlook · Human approvals

Prototype on **demo data** for cohort APD2F2606SE · Jan–Apr 2027 semester

github.com/joseph-mayala/crunchradar